Bayesian Classifier and its variants

Chapter 02

Ref. Book: Pattern Recognition – S. Theodoridis

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Classifier based on Bayes Decision theory

Statistical nature of feature vectors

$$\underline{x} = \left[x_1, x_2, \ldots, x_l\right]^T$$

Assign the pattern represented by feature vector to the most probable of the available classes

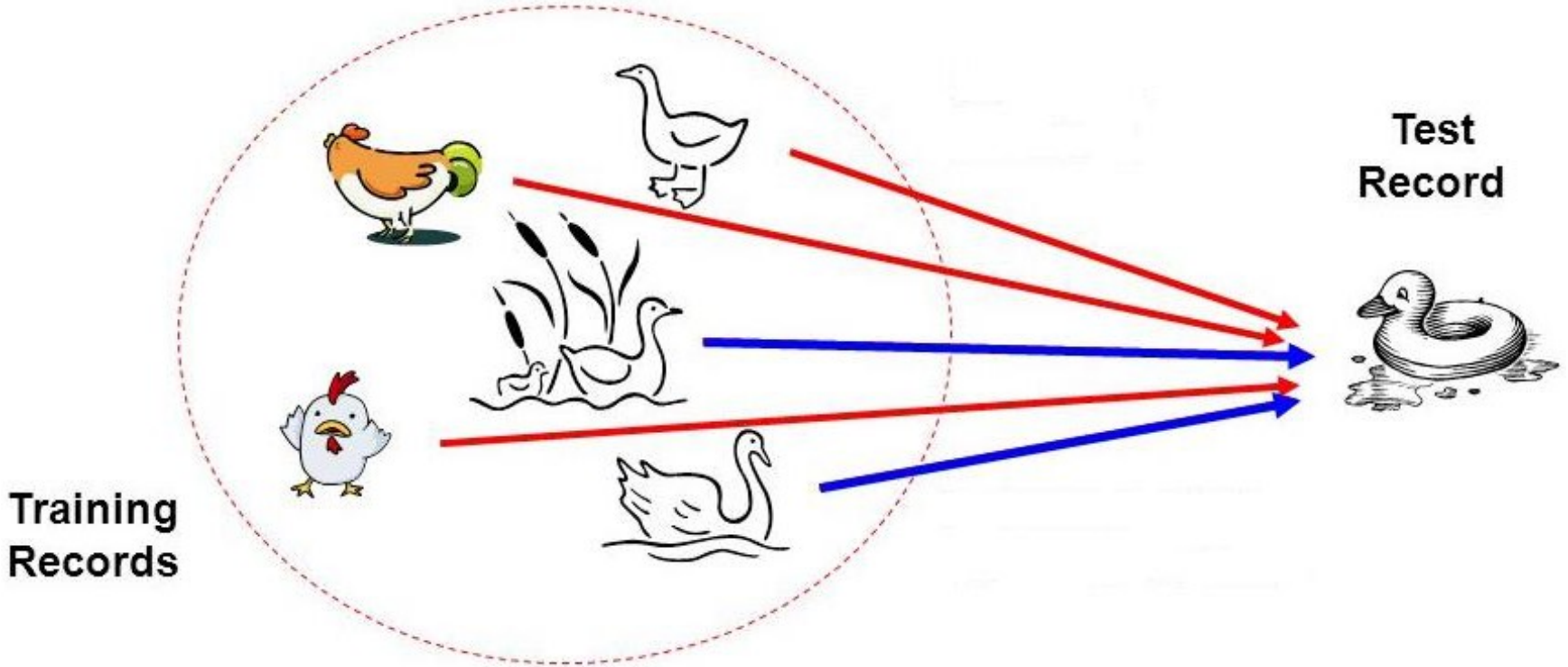
$$\omega_1, \omega_2, \ldots, \omega_M$$

That is $\underline{x} \rightarrow \omega_i : P(\omega_i \mid \underline{x})$ Maximum

Bayesian Classifier

Principle

If it walks like a duck, quacks like a duck, then it is probably a duck



Bayesian Classifier

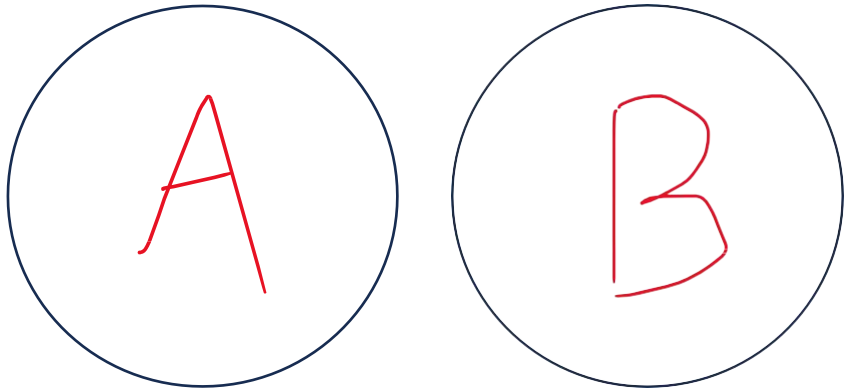
- A statistical classifier
  - Performs probabilistic prediction, i.e., predicts class membership probabilities
- Foundation
  - Based on Bayes' Theorem.
- Assumptions
  - The classes are mutually exclusive and exhaustive.
  - The attributes are independent given the class.
- Called "Naïve" classifier because of these assumptions
  - Empirically proven to be useful.
  - Scales very well.

Mutually Exclusive and Exhaustive events

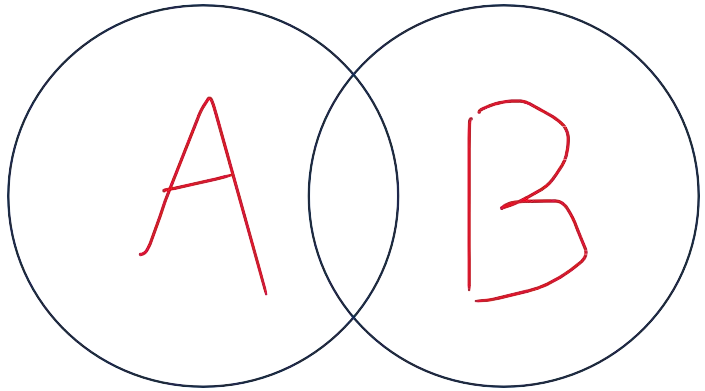
Mutually exclusive

Events they do not have any outcomes in common

- Disjoint
- $P(A \cap B)=0$



Mutually Exclusive



Mutually not Exclusive

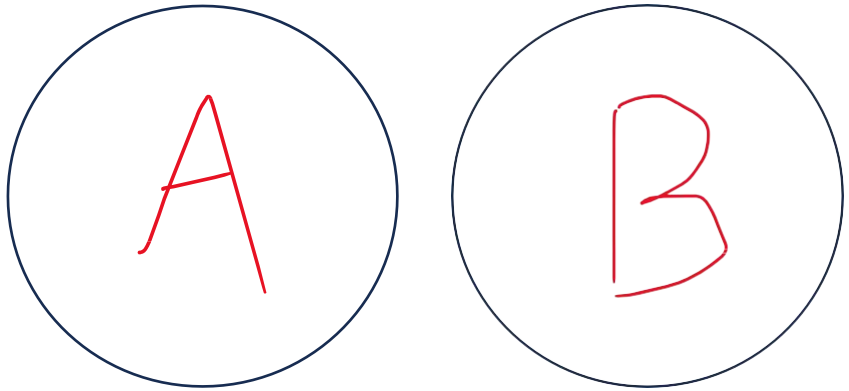
Mutually Exclusive and Exhaustive events

Mutually exclusive

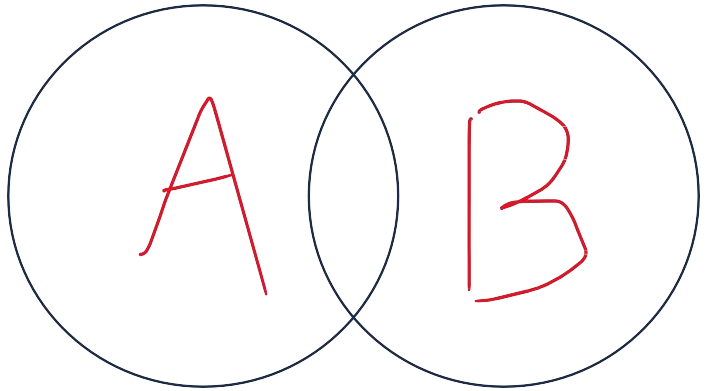
Event: Dice (36 sample spaces)

- Event A: P(at least one 5)
- Event B: P(sum<4)

Mutually exclusive?



Mutually Exclusive



Mutually not Exclusive

Mutually Exclusive and Exhaustive events

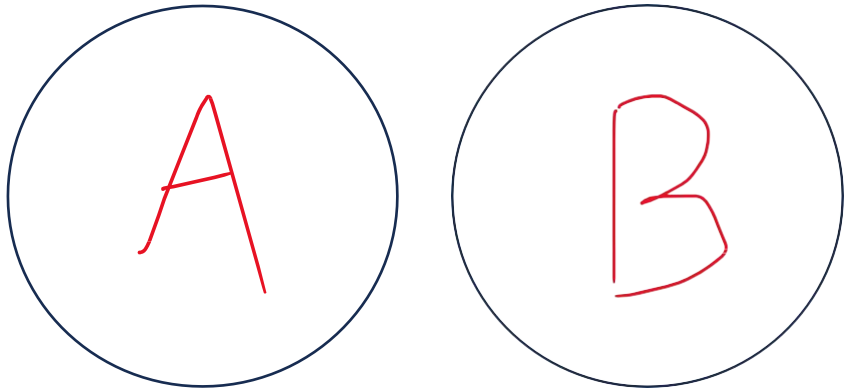
Mutually exclusive

Event: Dice (36 sample spaces)

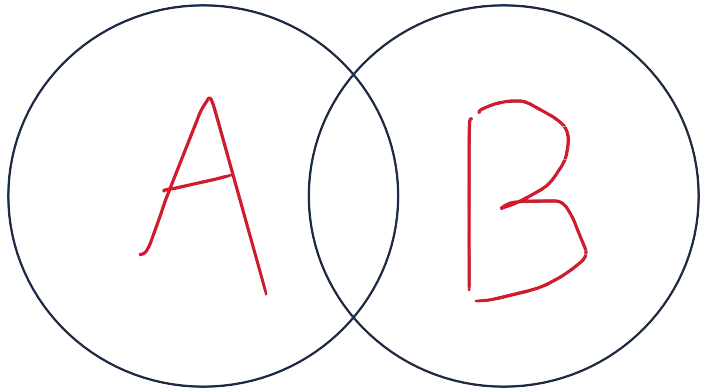
- Event A: P(at least one 5)
- Event B: P(sum<4)

YES

Mutually exclusive?



Mutually Exclusive



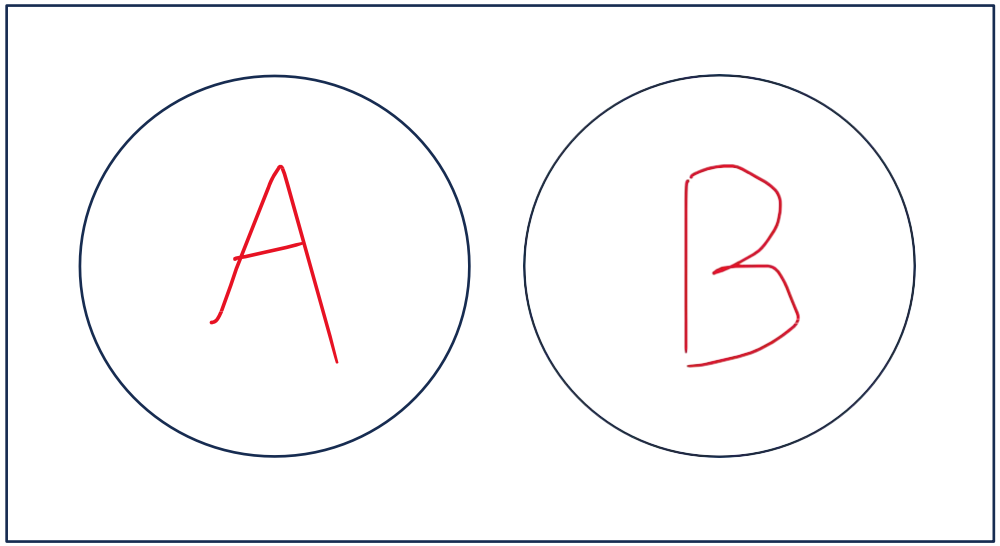
Mutually not Exclusive

Mutually Exclusive and Exhaustive events

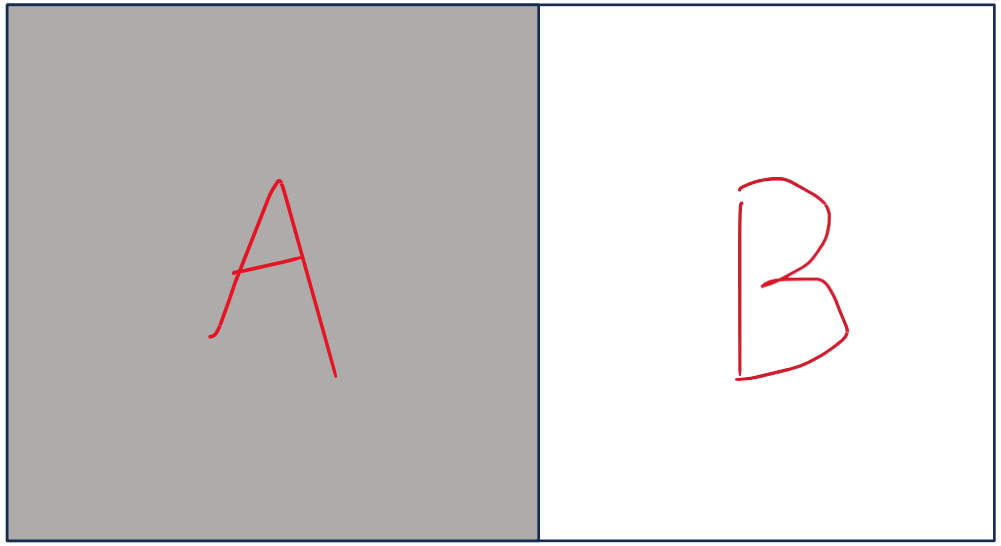
Exhaustive Events

Events that contain all the outcomes of sample space

- $P(A \cup B)=1$



Not Exhaustive



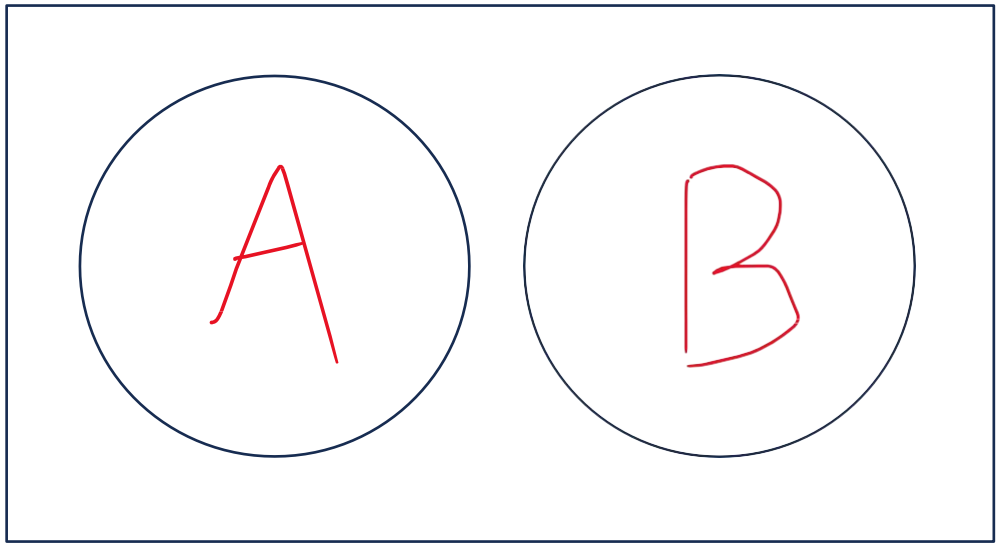
Exhaustive

Mutually Exclusive and Exhaustive events

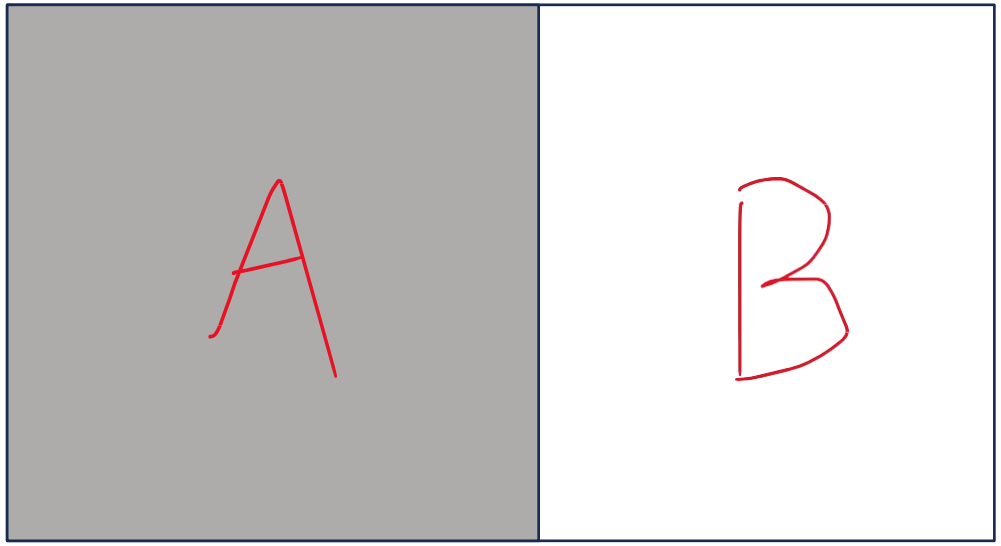
Exhaustive Events

Event: Dice (36 sample spaces)

- Event A: P(at least one 6)
- Event B: P(sum<11)



Not Exhaustive



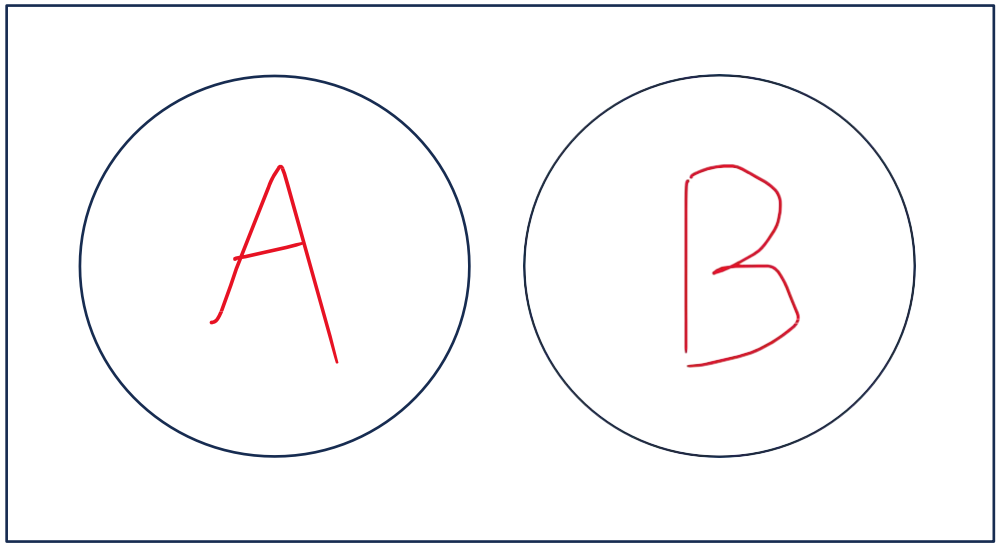
Exhaustive

Mutually Exclusive and Exhaustive events

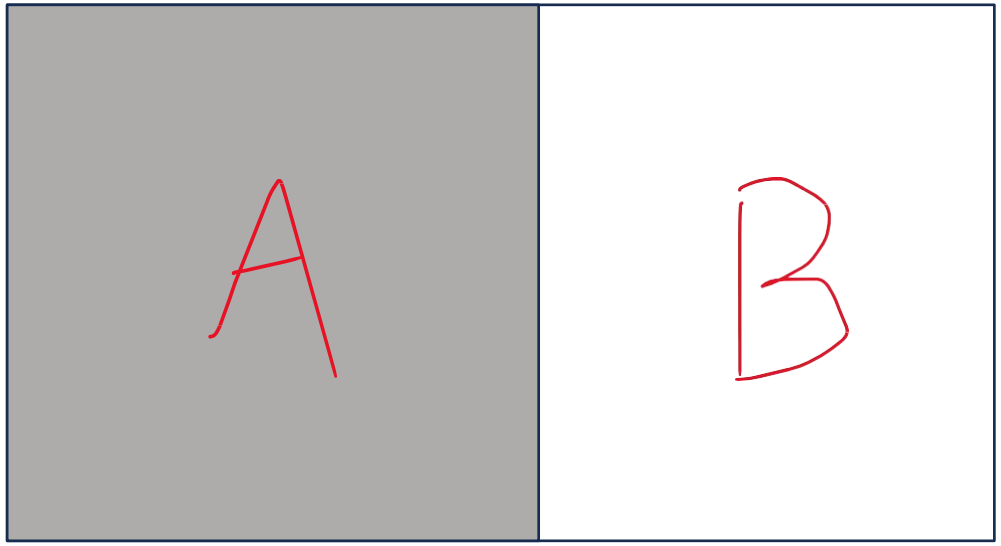
Mutually Exclusive + Exhaustive Events

When exhaustive events don't overlap, they become mutually exclusive!

Any example?



Not Exhaustive



Exhaustive

Bayesian Classifier Examples

Example 01

Given:

- A doctor knows that meningitis causes stiff neck 50% of the time
- Prior probability of any patient having meningitis is 1/50000
- Prior probability of any patient having stiff neck is 1/20

Question: If a patient has stiff neck, does he/she has meningitis?

Bayesian Classifier Examples

Example

- A married person with income 120k did not refund the loan previously
- Can we trust him?

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Bayesian Classifier Examples

Example 03

The sea bass/salmon example

- We now the previous counts of salmon/sea bass
- Can we predict which fish is coming in the conveyor?

Bayesian Classifier

- In many applications, the relationship between the attributes set and the class variable is non-deterministic.
  - In other words, a test cannot be classified to a class label with certainty.
  - In such a situation, the classification can be achieved probabilistically.

- The Bayesian classifier is an approach for modelling probabilistic relationships between the attribute set and the class variable.

- More precisely, Bayesian classifier use Bayes' Theorem of Probability for classification.

- Before going to discuss the Bayesian classifier, we should have a quick look at the Theory of Probability and then Bayes' Theorem.

Simple probability

If there are n elementary events associated with a random experiment and m of n of them are favorable to an event A, then the probability of happening or occurrence of A is

$$P(A) = \frac{m}{n}$$

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$$P(A) = \frac{m}{n}$$

Suppose, A and B are any two events and P(A), P(B) denote the probabilities that the events A and B will occur, respectively.

Mutually Exclusive Events: Two events are mutually exclusive, if the occurrence of one precludes the occurrence of the other.

Example: Tossing a coin (two events)
Tossing a ludo cube (Six events)

Can you give an example, so that two events are not mutually exclusive?

Hint: Tossing two identical coins, Weather (sunny, warm)

Simple probability

Independent events: Two events are independent if occurrences of one does not alter the occurrence of other.

Example: Tossing both coin and ludo cube together.
(How many events are here?)

- Can you give an example, where an event is dependent on one or more other events(s)?

  Hint: Receiving a message (A) through a communication channel (B) over a computer (C), rain and dating.

Joint probability

If P(A) and P(B) are the probability of two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive, then $P(A \cap B) = 0$

If A and B are independent events, then $P(A \cap B) = P(A) . P(B)$

Thus, for mutually exclusive events

$$P(A \cup B) = P(A) + P(B)$$

Conditional probability

If events are dependent, thentheir probability is expressed by conditional probability. The probability that A occurs given that B is denoted by $P(A|B)$.

Suppose, A and B are two events associated with a random experiment. The probability of A under the condition that B has already occurred and $P(B) \neq 0$ is given by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Bayes Classifier

A Probabilistic framework for solving classification problems

Conditional probability: $P(C|A) = \frac{P(A,C)}{P(A)}$

$(C A$

Bayes theorem:

$$P(C|A) = \frac{P(A|C)P(C)}{P(A)}$$

The End